

Question 1. Use a SEPARATE Writing Booklet

Marks

- (a) Find $\int 3x \sec^2(x^2) dx$ [2]
- (b) Find $\int \frac{e^{-x}}{\sqrt{1-e^{-x}}} dx$ [2]
- (c) Find $\int \frac{\tan^6 x}{\sin^5 x} dx$ [2]
- (d) Find $\int \cos^{-1} \theta \, d\theta$ [3]
- (e) Let $I_n = \int_0^\pi x^n \sin x dx$ where n is an integer
- (i) Show that $I_n = \pi^n - n(n-1)I_{n-2}$ for $n \geq 2$ [3]
- (ii) Hence evaluate I_5 [3]

Question 2

Use a SEPARATE Writing Booklet

Marks

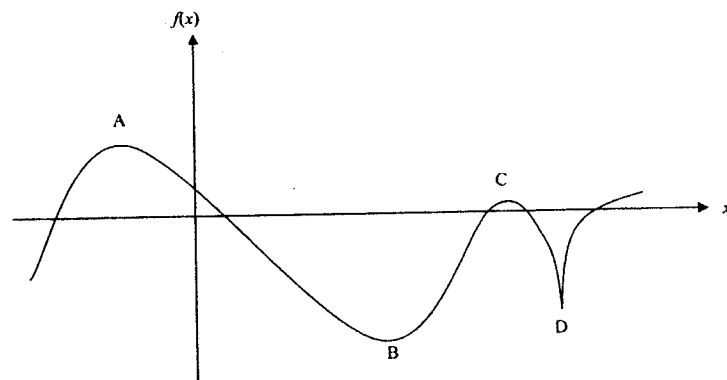
- (a) Given that $W = \frac{3+i}{1-2i}$, express the following in the form $a+ib$
- (i) $|W|$ [2]
- (ii) $\overline{W}$ [1]
- (iii) W^{-1} [1]
- (b) Illustrate with a diagram and describe in Geometric terms the locus, represented by the following:
- (i) $|i-z|=3$ [2]
- (ii) $\frac{\overline{z}z}{2} = \overline{z} + z$ [2]
- (c) Let z be a complex number such that $\arg(z)=\theta$, where $\frac{\pi}{2} < \theta < \pi$, and $|z|=1$
- Sketch z^2 and z on an Argand Diagram and find in terms of θ the values of
- (i) $\left| \frac{2}{1+z^2} \right|$ [3]
- (ii) $\arg\left(\frac{2}{1+z^2}\right)$ [4]

Question 3. Use a SEPARATE Writing Booklet

Marks

- (a) On your answer page make a rough copy of the sketch of $f(x)$ below. Label the turning points A , B , C and label the cusp D as indicated. On the same set of axes sketch the derivative function, $f'(x)$

[3]



- (b) Sketch the curves on two separate diagrams, show the equations of any asymptotes, and show any intercepts on the axes.

(i) $y = \frac{5-x}{x}$ [2]

(ii) $y = \frac{25-x^2}{x^2}$ [2]

Hence or otherwise sketch on another two separate diagrams, the curves

(iii) $y = \left| \frac{5-x}{x} \right|$ [2]

(iv) $y^2 = \frac{25-x^2}{x^2}$ [3]

- (c) Sketch and label $y = \sin^3(2x)$ for $0 \leq x \leq \pi$ and on the same axes sketch and label $y = \ln(\sin^3(2x))$ for $0 \leq x \leq \pi$.

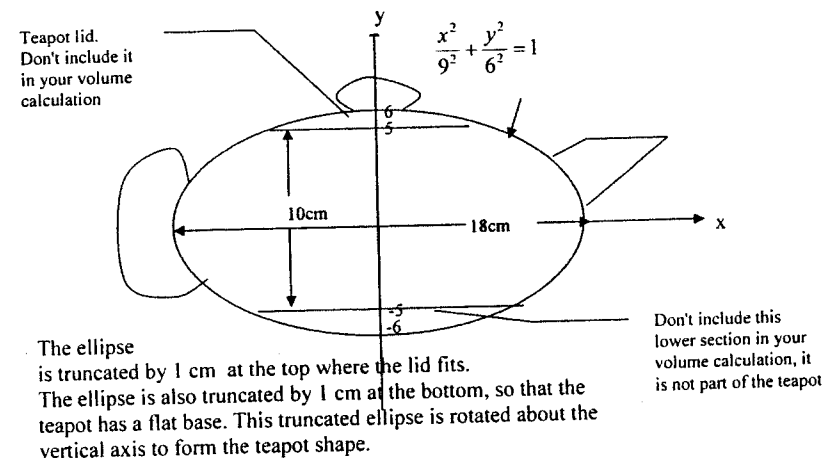
[3]

Question 4. Use a SEPARATE Writing Booklet

Marks

- (a) A large shiny metal teapot appears to be circular when viewed from above. The same teapot appears to be elliptical when viewed from the side (as shown).

[6]



Use the method of shells to find the volume of this shiny metal teapot and then express its capacity in litres correct to 2 decimal places.

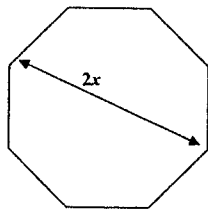
- (b) With the aid of a sketch and careful integration show that the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab square units.

[3]

Question 4 continues next page

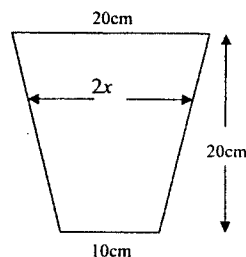
Question 4. continued

- (c) The horizontal cross-section of a vase is a regular octagon. The maximum width $2x$ of the octagonal cross-section is 10cm at the base and 20cm at the top.



The horizontal cross-section is shown opposite

The vertical cross-section is shown opposite.



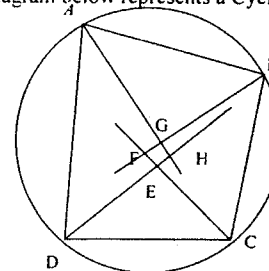
- (i) Find an expression for the area of the horizontal cross-section when the maximum width is $2x$ [2]
- (ii) Find the volume of the vase using the method of parallel cross-sections. [leave your answer in cm^3] [4]

Question 5. Use a SEPARATE Writing Booklet

Marks

- (a) A square-based pyramid, of base length, s and perpendicular height, h is inscribed in a sphere of radius r .
- (i) Show that $s^2 = 4hr - 2h^2$ [2]
- (ii) Find h in terms of r so that the pyramid has maximum volume. [3]
- (iii) Find an expression for the maximum volume in terms of r . [2]

- (b) The diagram below represents a Cyclic Quadrilateral $ABCD$.



Bisectors of the angles have been drawn, forming a smaller quadrilateral $EFGH$. Copy the diagram onto your answer page and Prove that $EFGH$ is also a Cyclic Quadrilateral [5]

- (c) Solve for x
 $\tan^{-1} 5x - \tan^{-1} 3x = \tan^{-1} \frac{1}{4}$ [3]

Question 6.	Use a SEPARATE Writing Booklet	Marks
(a).	(i) Express the complex cube roots of unity, ω and ω^2 in the form $re^{i\theta}$	[1]
	(ii) Using Argand diagram or otherwise, show that $\omega + \omega^2 + \omega^3 = 0$	[3]
	(iii) Prove that if $P(x) = p_0 + p_1x + p_2x^2 + \dots + p_nx^n$ has a root of multiplicity m , then $P'(x)$ has a root of multiplicity $(m-1)$.	[3]
	(iv) Prove that ω , a complex cube root of unity is a repeated root of the polynomial $P(x) = 5x^5 + 7x^4 + 9x^3 + x^2 - x - 3$	[4]
(b).	If α, β, γ are the roots of $x^3 - 9x + 9 = 0$, show that $(\alpha-1)(\beta-1)(\gamma-1) = -1$	[4]

Question 7.	Use a SEPARATE Writing Booklet	Marks
(a)	(i) Write down the parametric equations that correspond to the Cartesian equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	[1]
	(ii) Show that the greatest area of a rectangle inscribed in an ellipse is $2ab$ u ²	[2]
(b)	For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ find	
	(i) Eccentricity	[1]
	(ii) Coordinates of the foci	[1]
	(iii) Equations of the directrices	[1]
	(iv) Equation of the tangent to the hyperbola at the point $(5, -2\frac{1}{4})$	[2]
	(v) Show foci, directrices, asymptotes and the tangent on a neat sketch.	[4]
(c)	An hyperbola has asymptotes $y = \pm x$ and it passes through the point $(-3, -2)$.	
	(i) Find the equation of the hyperbola	[1]
	(ii) Find the length of the transverse axis	[1]
	(iii) Explain why this hyperbola is rectangular	[1]

Question 8. Use a SEPARATE Writing Booklet

Marks

(a) The positive integers are bracketed as follows:

(1), (2,3), (4,5,6),

where there are r integers in the r^{th} bracket.

Prove that the sum of the integers in the r^{th} bracket is $\frac{1}{2}r(r^2 + 1)$.

[5]

(b) If $a > 0, b > 0, c > 0$, show that $a^2 + b^2 + c^2 \geq ab + bc + ca$

and state the condition of equality.

[3]

(c) (i) Express $1 + x + x^2 + x^3 + x^4 + x^5$ as a product of real factors

[1]

(ii) Prove that the equation $c + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \frac{x^6}{6} = 0$

has no real roots if $c > \frac{37}{60}$

[6]

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

$$\begin{aligned} \text{Q1 a)} &= \frac{3}{2} \int 2x \sec^2(x^2) dx \\ &= \frac{3}{2} \int \sec^2(x^2) dx^2 \\ &= \frac{3}{2} \tan(x^2) + C \end{aligned}$$

$$\begin{aligned} \text{b)} &= \int (1-e^{-x})^{\frac{1}{2}} d(1-e^{-x}) \\ &= 2(1-e^{-x})^{\frac{3}{2}} + C \end{aligned}$$

$$\begin{aligned} \text{c)} &= \int \frac{\sin x}{\cos^5 x} dx \\ &= -\int \cos^{-5} x d \cos x \\ &= -\frac{\cos^{-5} x}{-5} \\ &= \frac{1}{5 \cos^5 x} + C \end{aligned}$$

$$\text{d)} \int \cos^{-1} \theta d\theta =$$

$$\begin{aligned} &= \theta \cdot \cos^{-1} \theta - \int \theta \cdot \frac{-1}{\sqrt{1-\theta^2}} d\theta \\ &= \theta \cdot \cos^{-1} \theta - \frac{1}{2} \int (1-\theta^2)^{-\frac{1}{2}} d(1-\theta^2) \\ &= \theta \cdot \cos^{-1} \theta - \frac{1}{2} \cdot \frac{(1-\theta^2)^{\frac{1}{2}}}{\frac{1}{2}} \\ &= \theta \cdot \cos^{-1} \theta - \sqrt{1-\theta^2} + C \end{aligned}$$

$$\begin{aligned} \text{e)(i)} I_n &= \int_0^{\pi} x^n \sin x dx \\ \therefore I_n &= \int_0^{\pi} x^n d \cos x \end{aligned}$$

$$= [-x^n \cos x]_0^{\pi} - \int_0^{\pi} \cos x \cdot -n x^{n-1} dx$$

$$= (-x^n \cdot -1) + \int_0^{\pi} n x^{n-1} d \sin x$$

$$= x^n + [n x^{n-1} \sin x]_0^{\pi} - \int_0^{\pi} \sin x \cdot n(n-1) x^{n-2} dx$$

$$= x^n - n(n-1) I_{n-2} \quad \text{for } n \geq 2$$

Q1 cont.

$$\begin{aligned} \text{e(ii)} I_5 &= \pi^5 - 20 I_3 \\ &= \pi^5 - 20(\pi^3 - 6 I_1) \\ &= \pi^5 - 20\pi^3 + 120 \int_0^{\pi} x \sin x dx \end{aligned}$$

$$\begin{aligned} \text{but } \int x \sin x dx &= \int x d(-\cos x) \\ &= [-x \cos x]_0^{\pi} + \int_0^{\pi} \cos x dx \\ &= \pi + [\sin x]_0^{\pi} \\ &= \pi \end{aligned}$$

$$\therefore I_5 = \pi^5 - 20\pi^3 + 120\pi$$

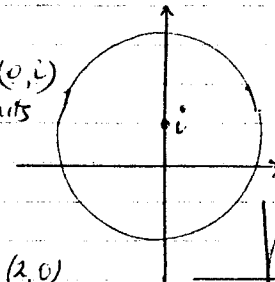
$$\begin{aligned} \text{Q2 a)(i)} W &= \frac{3+i}{1-2i} \times \frac{1+2i}{1+2i} \\ &= \frac{1+7i}{5} \end{aligned}$$

$$\therefore |W| = \sqrt{\frac{1}{25} + \frac{49}{25}} = \sqrt{2}$$

$$\text{(ii)} \bar{W} = \frac{1}{5} - \frac{7i}{5}$$

$$\begin{aligned} \text{(iii)} \frac{1}{W} &= \frac{\bar{W}}{W\bar{W}} \\ &= \frac{\bar{W}}{|W|^2} \\ &= \frac{1-7i}{5} \times \frac{1}{(\sqrt{2})^2} \\ &= \frac{1}{10} - \frac{7i}{10} \end{aligned}$$

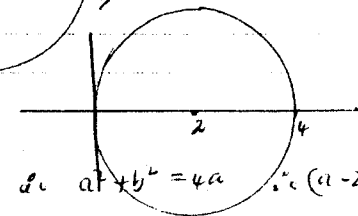
b)(i)
circle centre (0, i)
radius 3 units



(ii)
circle centre (2, 0)
radius 2.

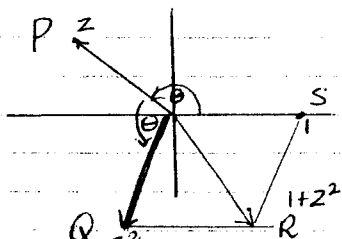
$$\text{Working: } a^2 + b^2 = a^2 + 0 + 0 + 0$$

$$\therefore a^2 + b^2 = 4 \quad \therefore (a-2)^2 + b^2 = 4$$



Q2 ctd.

c)



algebraic method

$$\begin{aligned} \text{(i)} \quad 1+z^2 &= 1 + \cos 2\theta + i \sin 2\theta \\ \therefore |1+z^2| &= \sqrt{1 + 2\cos 2\theta + \cos^2 2\theta + \sin^2 2\theta} \\ &= \sqrt{2 + 2\cos 2\theta} \\ &= \sqrt{2(1 + \cos 2\theta)} \\ &= \sqrt{4 \cos^2 \theta} \\ &= 2 \cos \theta \end{aligned}$$

$$\therefore \left| \frac{2}{1+z^2} \right| = \frac{|2|}{|2 \cos \theta|} = \sec \theta$$

geometric method

OQRS is a rhombus, side length 1 unit.

 $\angle OS = 360^\circ - 2\theta$ (angles at a point). $\angle OR = 180^\circ - (360^\circ - 2\theta)$ (co-interior angles, $OS \parallel QR$)

$$= 2\theta - 180^\circ$$

In $\triangle OQR$

$$OR = |1+z^2|$$

$$= 2$$

$$\text{and } q^2 = r^2 + o^2 - 2 \times r \times o \times \cos(2\theta - 180^\circ)$$

$$= 1 + 1 - 2(\cos 2\theta \cos 180^\circ + \sin 2\theta \sin 180^\circ)$$

$$= 2 - 2(-\cos 2\theta)$$

$$= 2 + 2\cos 2\theta$$

$$= 4 \cos^2 \theta$$

$$\therefore q = 2 \cos \theta$$

$$\therefore \left| \frac{2}{1+z^2} \right| = \frac{2}{2 \cos \theta}$$

$$= \sec \theta$$

Q2 ctd.

geometric method.

c(ii) OR bisects $\angle OS$

$$= \frac{1}{2} (2\pi - 2\theta) = \pi - \theta$$

$$\therefore \arg(1+z^2) = -(\pi - \theta)$$

$$\begin{aligned} \text{Now } \arg\left(\frac{2}{1+z^2}\right) &= \arg 2 - \arg(1+z^2) \\ &= 0 - (\theta - \pi) \\ &= \pi - \theta \end{aligned}$$

or algebraic method

$$\begin{aligned} \arg\left(\frac{2}{1+z^2}\right) &= \arg 2 - \arg(1+z^2) \\ &= 0 - \arg(1 + \cos 2\theta + i \sin 2\theta) \\ &= -\tan^{-1}\left(\frac{\sin 2\theta}{1 + \cos 2\theta}\right) \end{aligned}$$

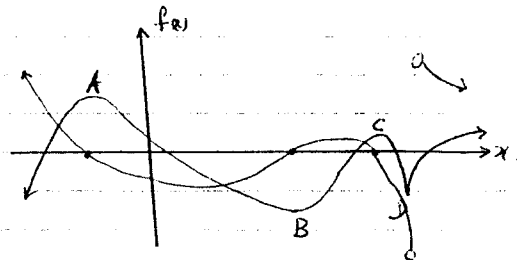
$$= -\tan^{-1}\left(\frac{2 \sin \theta \cos \theta}{1 + 2 \cos^2 \theta - 1}\right)$$

$$= -\tan^{-1}(\tan \theta)$$

but θ is in 2nd quadrant and $\tan^{-1} x$ is defined: $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$

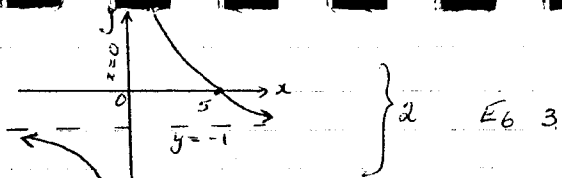
$$\begin{aligned} \therefore \arg\left(\frac{2}{1+z^2}\right) &= -\tan^{-1}(-\tan(\pi - \theta)) \\ &= -\tan^{-1} \tan(\theta - \pi) \\ &= -(\theta - \pi) \\ &= \pi - \theta \end{aligned}$$

Q3 a)

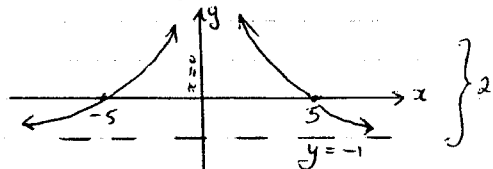


Q3 c) 1

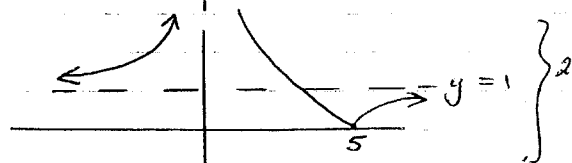
b) (i) $y = \frac{x}{2} - 1$



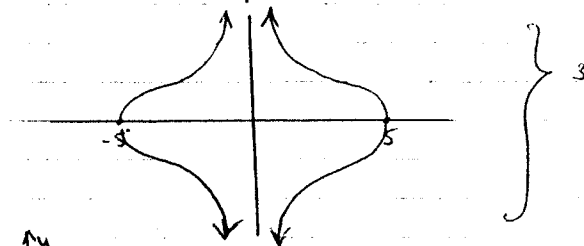
(ii) $y = \frac{25}{x^2} - 1$



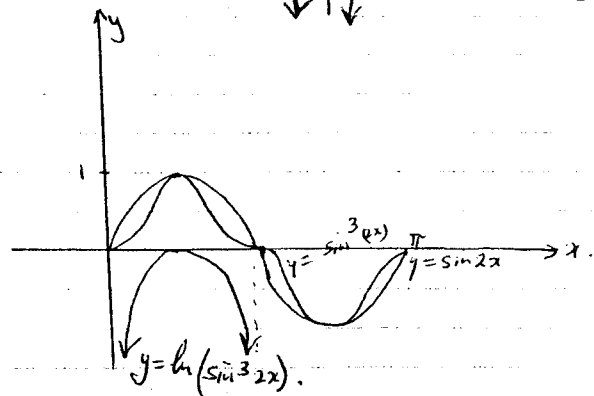
(iii)



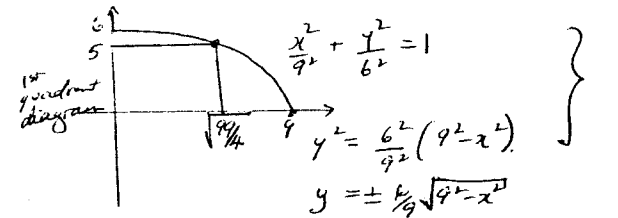
(iv)



c)



Q4 a)



Typical shell

$\delta V = 2\pi xy \delta x$

Total Volume

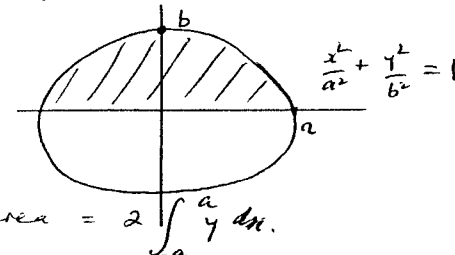
$$= 2\pi \left[\int_0^{\sqrt{9/4}} 5x dx + \frac{6}{9} \int_{\sqrt{9/4}}^9 x \cdot \sqrt{9^2 - x^2} dx \right]$$

$$= 4\pi \left[\left[\frac{5x^2}{2} \right]_0^{\sqrt{9/4}} - \frac{1}{2} \cdot \frac{2}{3} \left[(9^2 - x^2)^{3/2} \right]_{\sqrt{9/4}}^9 \right]$$

$$= 4\pi \left(\frac{5 \times 9}{2} - \frac{2(-225)^{3/2}}{9} \right)$$

$$= 1955.64 \dots \text{cm}^3 \text{ or ml}$$

$$\approx 1.96 \text{ Litres}$$



Total area $= 2 \int_a^b y dx$

where $y^2 = \frac{b^2}{a^2} (a^2 - x^2)$

$$\therefore \text{Area} = 2 \cdot \frac{b}{a} \int_a^b \sqrt{a^2 - x^2} dx$$

but $\int_a^b \sqrt{a^2 - x^2} dx$ is a semicircular area

$$= \frac{1}{2} \pi a^2$$

$$\therefore \text{Area of ellipse} = 2 \times \frac{b}{a} \times \frac{1}{2} \pi a^2$$

$$= \pi ab \text{ u}^2$$

Q4 c)

(i) 8 sections

$$\text{Area} = \frac{1}{2} x^2 \sin 45^\circ$$

$$= \frac{1}{2} x^2 \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{2} x^2}{4}$$

$$\therefore \text{Total cross-section} = 8 \times \frac{\sqrt{2} x^2}{4} = 2\sqrt{2} x^2 \text{ m}^2$$

$$(ii) SV = Agh$$

$$= 2\sqrt{2} x^2 gh$$

$$\text{Now } 2x = \frac{h}{2} + 10 \text{ where } 0 \leq h \leq 20$$

$$\therefore x = \frac{h}{4} + \frac{20}{4}$$

$$\therefore V = 2\sqrt{2} \int_0^{20} \left(\frac{h+20}{4} \right)^2 dh$$

$$= \frac{2\sqrt{2}}{16} \int_0^{20} (h^2 + 40h + 400) dh$$

$$= \frac{\sqrt{2}}{8} \left[\frac{h^3}{3} + 20h^2 + 400h \right]_0^{20}$$

$$= \frac{\sqrt{2}}{8} (8000 + 8000 + 8000)$$

$$= \frac{\sqrt{2}}{8} \times \frac{56000}{3}$$

$$= \frac{7000\sqrt{2}}{3}$$

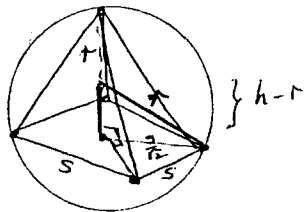
$$\approx 3360 \text{ cm}^3$$

2. E72

$$Q5a) i) (h-r)^2 + \left(\frac{s}{2}\right)^2 = r^2$$

$$\frac{s^2}{4} + h^2 - 2hr + r^2 = r^2$$

$$s^2 = 4hr - 2h^2$$



$$(ii) \text{Area of base} = s^2$$

$$\text{Vol of pyramid} = \frac{1}{3} s^2 h$$

$$= \frac{1}{3} (4hr - 2h^2) h$$

$$\frac{dV}{dh} = \frac{1}{3} (8hr - 6h^2) = \frac{2h}{3} (4r - 3h)$$

$$= 0 \text{ for } h = \frac{4r}{3}$$

Q5 a) ctd

(i)

$$\frac{d^2V}{dh^2} = \frac{8r - 12h}{3}$$

$$\text{and at } h = \frac{4r}{3}$$

$$= -\frac{8r}{3}$$

$$< 0 \therefore \text{max volume when } h = \frac{4r}{3}$$

3

E64

$$(ii) \therefore \text{Max Vol} = \frac{1}{3} \left(4 \times \frac{16r^2}{9} \cdot r - 2 \times \frac{64r^3}{27} \right)$$

$$= \frac{1}{3} \left(\frac{192 - 128}{27} r^3 \right)$$

$$= \frac{64r^3}{81}$$

u3

2. E64

b) proof

$$\text{Let } \angle ABC = 2x \text{ (opp. } \angle \text{ s of cyclic quad)}$$

$$\therefore \angle ADC = 180 - 2x$$

$$\text{Similarly } \angle ADB = 2y$$

$$\text{In } EFGH$$

$$\angle FGH = 180 - (90 - x + 90 - y) = x + y$$

$$\text{and } \angle FEH = 180 - (x + y) \text{ (sum of } \angle \text{ s in } \triangle AEB)$$

$$\therefore EFGH \text{ is a cyclic quad}$$

$$\text{Since } \angle FGH + \angle FEH = 180 \text{ (opp } \angle \text{ s supplementary)}$$

5 E1,2

$$c) \tan(\tan^{-1} 5x - \tan^{-1} 3x) = \tan(\tan^{-1} \frac{1}{7})$$

$$\text{LHS} = \frac{5x - 3x}{1 + 15x^2} \quad \text{RHS} = \frac{1}{7}$$

$$\therefore 8x = 1 + 15x^2$$

$$(5x - 1)(3x - 1) = 0$$

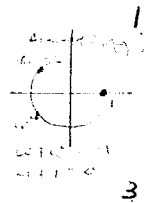
$$x = \frac{1}{5}, \frac{1}{3}$$

3 E4, 1

Q6 a) (i) $\omega = \cos 2\pi/3$
 $\omega^2 = \cos 4\pi/3$

$\omega^3 = \cos 6\pi/3 = \cos 2\pi = 1$

(ii) $\omega + \omega^2 + \omega^3 = -\frac{1}{2} + \frac{j\sqrt{3}}{2} + \frac{1}{2} - \frac{j\sqrt{3}}{2} + 1$
 $= -1 + 1$
 $= 0$



E₃ 2

E₃ 2

(iii) Let $P(x) = (x-\alpha)^m \cdot Q(x)$

where $Q(\alpha) \neq 0$

Then $P'(x) = m(x-\alpha)^{m-1} Q(x) + Q'(\alpha) \cdot (x-\alpha)^m$
 $= (x-\alpha)^{m-1} (mQ(\alpha) + Q'(\alpha)(x-\alpha))$
 $= (x-\alpha)^{m-1} R(x)$

$\therefore (x-\alpha)$ is a factor of $P'(x)$ and
 α is a root of $P'(x) = 0$ of order $m-1$.

3.

E₃ 3.

(iv) If ω is a repeated root

then $P(\omega) = P'(\omega) = 0$

now $P(x) = 25x^4 + 28x^3 + 27x^2 + 2x - 1$

$\therefore P'(\omega) = 25 \times 4\omega + 28 \times 3\omega^2 + 27 \times 2\omega + 2 = 0$

$= 27\omega + 27\omega^2 + 27$

$= 27(\omega + \omega^2 + 1)$ from (ii) above

$= 27 \times 0$

$= 0$

4

E₃ 2

b) Since α, β, γ are roots then

$\alpha^3 - 9\alpha + 9 = 0 \quad \text{i.e. } \alpha^3 = 9(\alpha - 1)$

$\beta^3 - 9\beta + 9 = 0 \quad \beta^3 = 9(\beta - 1)$

$\gamma^3 - 9\gamma + 9 = 0 \quad \gamma^3 = 9(\gamma - 1)$

$\therefore \alpha^3 \beta^3 \gamma^3 = 9^3 (\alpha - 1)(\beta - 1)(\gamma - 1)$

but $\alpha\beta\gamma = -9$

$\therefore (-9)^3 = -81 \quad \therefore 81(\alpha - 1)(\beta - 1)(\gamma - 1)$

$\therefore (\alpha - 1)(\beta - 1)(\gamma - 1) = -1$ as required

4

E₃ 3.

all the roots of $x^3 + 3x^2 - 6x + 1 = 0$ are real

roots are α, β, γ $\therefore \alpha^3 + 3\alpha^2 - 6\alpha + 1 = 0$ 1 point each

Q7.

a) (i) $x = a \cos \theta \quad y = b \sin \theta$

(ii) area of rectangle $\{ = 2x \times 2y$

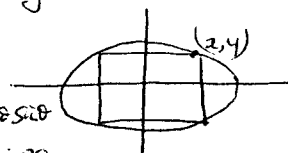
$= 4ab \cos \theta \sin \theta$

$= 2ab \sin 2\theta$

$= 2ab$ when $\sin 2\theta = 1$

$2\theta = \pi/2$

$\theta = \pi/4$



b) (i) $b^2 = a^2(e^2 - 1)$

$e^2 = \frac{b^2}{a^2} + 1$

$= \frac{9}{16} + 1$

$= 25/16$

$\therefore e = 5/4 \quad (e > 0)$

(ii) foci $(\pm ae, 0)$ i.e. $(\pm 5, 0)$

(iii) $x = \pm a/e$ i.e. $x = \pm 16/5$

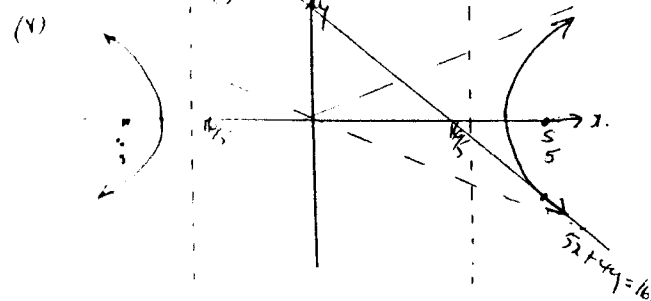
(iv) $\frac{\partial x}{\partial t} = \frac{\partial y}{\partial t} \cdot \frac{dy}{dx} = 0$

$\frac{dx}{dt} = \frac{9x}{16y}$

$= \frac{-5}{4}$

if $\frac{dx}{dt} = 0$ then $y = \frac{9}{4}(x-5)$

$5x + 4y = 16$



Q7

$$c) (i) y = \pm x \Rightarrow a = b = 1$$

$$\therefore \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

now $(-3, 2)$ satisfies

$$\therefore \frac{9}{a^2} - \frac{4}{b^2} = 1$$

$$\therefore 5 = a^2$$

$$\therefore \text{equation is } \frac{x^2}{5} - \frac{y^2}{5} = 1$$

note equation
can be expressed
as $xy = \frac{5}{2}$ (referred
to new axes)

1 Eq 3

$$(ii) 2\sqrt{5}$$

(iii) because asymptotes
are perpendicular to each other

Q8 By inspection it can be seen that
the last digit is a triangular number

$$1 \quad 1+2 \quad 1+2+3 \quad 1+2+3+4 \quad \dots$$

summing these, $a=1$, $d=1$ for each
bracket $\therefore$ the last digit in the
 $(r-1)^{\text{th}}$ bracket is

$$1+2+3+\dots+(r-1) = \frac{r-1}{2} (1+r-1) \\ = \frac{r-1}{2} \cdot r$$

now the next consecutive number

ie $\left(\frac{r-1}{2} \cdot r + 1\right)$ is the 1st digit in the r^{th} bracket.

But the last digit in the r^{th} bracket

$$= \frac{r}{2} (1+r)$$

$\therefore$ Sum of integers in r^{th} bracket

$$= \frac{r}{2} \left(\frac{r-1}{2} \cdot r + 1 + \frac{r+1}{2} \cdot r \right)$$

$$= \frac{r}{2} \left(\frac{r}{2} (r+1) + 1 \right)$$

$$= \frac{r}{2} (r^2 + 1) \text{ as required.}$$

5. E₁₂₉ 4.

$$b) (a-b)^2 = a^2 + b^2 - 2ab$$

$$a^2 + b^2 \geq 2ab \quad \text{equality for } a=b$$

$$\text{Similarly } b^2 + c^2 \geq 2bc$$

$$c^2 + a^2 \geq 2ca$$

$$\therefore \frac{1}{2}(a^2 + b^2 + c^2) \geq \frac{1}{2}(ab + bc + ca) \text{ equality for } a=b=c$$

8c)

$$(i) = 1 + x + x^2(1+x) + x^4(1+x) \\ = (1+x)(1+x^2+x^4)$$

1 E₁₂₉ 4

$$(ii) \text{ Let } f(x) = \frac{x^6}{6} + \frac{x^5}{5} + \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x + c$$

$$\text{Then } f'(x) = x^5 + x^4 + x^3 + x^2 + x + 1$$

$= 0$ for stationary pts

$$\text{When } (1+x)(1+x^2+x^4) = 0$$

but $1+x^2+x^4$ has no real roots

since $x^2 \geq 0$, $x^4 \geq 0$

$$\therefore f'(x) = 0 \text{ only for } x = -1$$

also

$$f''(x) = 5x^4 + 4x^3 + 3x^2 + 2x + 1$$

$$\therefore f''(-1) = 5 - 4 + 3 - 2 + 1$$

$$= 3$$

$> 0 \therefore$ min^m T.P. when $x = -1$

$\therefore$ for no real roots, need $f(-1) > 0$

ie need

$$\frac{(-1)^6}{6} + \frac{(-1)^5}{5} + \frac{(-1)^4}{4} + \frac{(-1)^3}{3} + \frac{(-1)^2}{2} + -1 + c > 0$$

$$\text{ie } \frac{1}{6} - \frac{1}{5} + \frac{1}{4} - \frac{1}{3} + \frac{1}{2} - 1 + c > 0$$

$$\frac{10 - 12 + 15 - 20 + 30 - 60}{60} + c > 0$$

$$-\frac{37}{60} + c > 0$$

$$\therefore c > \frac{37}{60}$$

6 " " "